# CaneVision: Attachable Navigation Module for White Cane

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**Introduction to Electronics, Microcontrollers, and Robotics**

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## Executive Summary

Visually impaired individuals face significant daily navigational challenges, particularly in detecting the upper-body obstacles and sudden drop-offs which the traditional white canes cannot sense. The report focuses on the design, development, testing, and application of the CaneVision, an attachable navigation module for white cane. Specifically, the product is a low-cost and attachable smart navigational module, aiming to upgrade the traditional white cane. Powered by Arduino Uno, the system integrates multiple sensors, including ultrasonic sensor for mid-air obstacle detection, infrared sensor for drop-offs avoidance and photoresistor for ambient light safety, to provide a comprehensive environmental detection. To ensure an intuitive interaction with the users to provide immediate feedback, the system utilizes the vibration motor for haptic feedback and buzzer for auditory alerts. After verifications and tests, the preliminary design validation achieves the initial expectations. Specifically, the components work interdependently, and the multi-sensor system successfully identifies the obstacle within the range from 0.8 m to 2.0 m. The feedback system provides alerts effectively and accurately when detecting obstacles. After real application, the product proves that the advanced, multi-direction spatial detection can be achieved affordably and effectively, enhancing independent safety for the visually impaired.

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## I. Introduction

Around the world, an estimated 2.2 billion people live with some form of vision impairment, among whom approximately 39 million are completely blind (World Health Organization [WHO], 2019). For the visually impaired individuals the spatial orientation and independent navigation present significance, which can directly impact their life quality and personal safety. For individuals, it is essential to perceive environmental obstacles in real time. For this reason, the travel aid devices are important for the visually impaired in both daily and unfamiliar environments.

Nowadays, the traditional primary white cane is still the widely adopted aid device for the visually impaired because of its simplicity and low cost. However, it suffers from some severe physical limitations. The limitations leave detection blind spots, making the users vulnerable to the hazards, which I will extend and make further explanation with detailed data in the Literature Review.

Considering the problems, the project presents CaneVision, an attachable aid module designed for the traditional white cane. The CaneVision is a light add-on part, not a full cane replacement. Arduino Uno works as the central microcontroller. To achieve the function of detection, I set a sensor group: an ultrasonic sensor for objects in the air, an IR sensor that detects ground drop-offs, and a photoresistor to detect the dark environment. For the actuators, the system sends vibration from a small motor to the user's hand and sound from a buzzer to the ear. In this way, the user can notice more

of the surroundings, while still using the cane in the usual way.

The purpose of this report is to document the complete engineering design process. The rest of this paper is organized as follows: Section 2 outlines the design requirements; Section 3 presents a review of existing mobility research and provides potential solutions; Section 4 focuses on the hardware and software design methodologies; Section 5 discusses verification procedures and test results; and the final sections offer conclusions and engineering recommendations for future improvements.

## II. Requirements

Aiming to address the limitations of existing products, I set some specific design requirements for the final design. These specific rules will help to maximize user safety and maintain overall affordability.

### i. Design Criteria

#### *a. Performance Criteria*

The primary functional requirement of the module is the reliable detection of objects at various levels. Specifically, to eliminate the blind zone left by traditional white canes, the device must accurately detect the upper-body obstacles positioned at heights between 0.8m and 2.0m, which is derived from the Americans with Disabilities Act (ADA) Standards for Accessible Design. The standards classify hazards positioned above standard detection height (0.68m) up to standard headroom clearance (2.03m) as critical collision risks (U.S. Department of Justice, 2010). In addition, the system establishes a horizontal forward detection range of 20 cm to 200

cm. Based on the standard average pedestrian walking speed of 1.2m/s (Federal Highway Administration [FHWA], 2009), a maximum detection boundary of 200 cm guarantees a preview time of approximately 1.5 seconds. In this way, the module can provide users with adequate perception-reaction time (Bhatlawande et al., 2014).

To show the distance from the user to an obstacle more clearly, a dual-zone alarm mode is used by the system. The "Warning Zone" (50 cm - 150 cm) has a reaction time of 0.4 - 1.25 seconds, and as the distance decreases, the frequency of intermittent feedback will increase. At the same time, the "Danger Zone" (below 50 cm) is a high-risk physical buffer area where the remaining reaction time is less than 0.4 seconds; this zone will trigger continuous and strong feedback, indicating that a stop is urgently required. In addition, the downward-facing infrared subsystem can detect a sudden drop in ground level to prevent falls (Pyun et al., 2013), and the combined photoresistor will automatically turn on the safety LED warning light in the event of low ambient light. In this way, the system maximizes the users' safety.

***b. Economic Constraints***

Existing commercial Electronic Travel Aids (ETA) are too expensive for the average users. For instance, according to the Lighthouse Guild (an internationally renowned non-profit organization for vision rehabilitation), current commercial visual assistive devices such as the Envision Glasses ($2,499 - $3,499) and eSight ($4,900+) require a high initial investment, which may be a huge economic obstacle for the visually impaired. Considering this circumstance, the economic constraints must be a main factor that we need to focus on. In this way, the proposed CaneVision module is

designed to achieve multi-zone hazard detection, keeping the total material cost well under $50, therefore maximizing the accessibility for the visually impaired community.

**ii. Design Specifications**

Beyond the sensory capabilities, the user interface of the module must not hinder the user's dexterity or their environmental awareness.

Considering the utility of the device, it must be designed to be a compact and lightweight attached module that secures firmly to any standard white cane. Mechanically, to preserve the basic function of sweeping and balance of the white cane, the total weight of the attached module must not exceed 150 grams. In addition, the mechanism must match standard white cane shaft diameters ranging from 13 mm to 16 mm. This range corresponds to the standard 1/2-inch to 5/8-inch outer diameter tubing used by industry-leading mobility cane manufacturers, such as Ambutech.

During active operation, the module must be unobtrusive and easily manipulated by the users. To report the obstacles without overwhelming the user's auditory senses, the system must prioritize haptic feedback (vibration). Additionally, this tactile alert is supplemented by a buzzer for critical danger warnings. Finally, the system must be entirely portable to ensure enough time for the users to use during the day.

In terms of the subsystem performance, the ultrasonic subsystem must maintain a forward detection range of 20 cm to 200 cm with an accuracy margin of ±5%. To guarantee adequate reaction time for the user walking at 1.2 m/s, the total system

latency must be under 100 milliseconds.

## III. Research and Literature Review

Autonomous navigation in unfamiliar environments is still a significant challenge for visually impaired individuals. According to Elmannai and Elleithy (2017), assistive devices can generally be categorized into simple electronic travel aids (ETAs), electronic orientation aids, and position locator devices. Their research indicates that in the field of electronic travel device, the primary engineering challenge is to balance detection range, response time, and user cognitive load.

Though human society experienced huge technological innovations, the traditional white cane remains the most widely adopted aid device for the visually impaired due to its affordability and intuitive nature. When studying low-cost electronic travelling aid devices, Ali (2023) discovered that while wearable smart belts provide multi-directional obstacle detection, smart canes are always rated higher in user's using feedback because they are close to the user's natural intuition. However, traditional white canes suffer from a significant deficit: they can primarily detect ground-level boundaries, leaving a great detection blind zone from the waist up to the user's head level. This makes users highly vulnerable to the upper-body obstacles, such as branches and signs on the roads. Finally, considering that the standard average pedestrian walking speed is approximately 1.2 m/s (Federal Highway Administration, 2009), traditional canes are physically incapable of providing preview warnings to create a buffer zone for the users facing mid-air

obstacles or sudden ground-level drop-offs until the cane tip touches the physical edge.

To provide a broader “view” for the visually impaired through detection assistive device, recent advancements have integrated multi-modal perception systems. For instance, Zhang et al. (2021) developed a unified perception architecture utilizing LiDAR cameras and AI-driven semantic segmentation to provide comprehensive scene recognition. Similarly, the MAPS system introduced by Romeo et al. (2022) utilizes pre-journey spatial learning combined with wearable "TactiBelt" feedback for global path awareness. Although these solutions are precise and mature, they are still too expensive for most of the ordinary users. In addition, complex ETAs often deliver continuous, high-bandwidth audio feedback. Research indicates that continuous auditory alerts can cause sensory overload, which dangerously interfere with a blind user's crucial ability to hear ambient environmental cues, such as approaching traffic (Bhatlawande et al., 2014).

Based on these findings and research, recent studies suggest that the most effective assistive devices are those that supplement, rather than replace, traditional orientation skills. Additionally, utilizing haptic feedback as the primary communication channel preserves the user's auditory channel and prevents cognitive overload (Romeo et al., 2022; Bhatlawande et al., 2014). In my design of CaneVision, this attachable navigational module for white canes provides multi-direction detection with a user-centric feedback design.

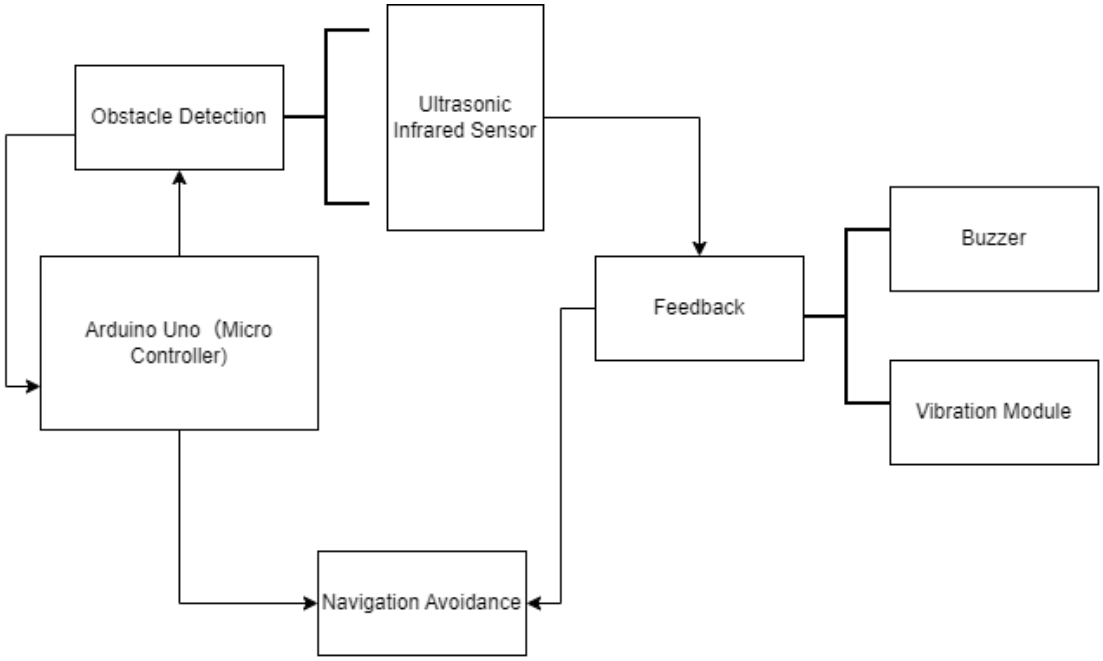
# IV. Design Method

## i. Operational Workflow

The CaneVision module can be divided into two primary subsystems: the hardware architecture and the software control logic. Specifically, considering the function of components, the module can be divided into three main functional subsystems: Central Processing Unit (Controller), the Sensing Subsystem (Obstacle Detector), and the Feedback Subsystem. The operational workflow among these modules is illustrated in Figure 1 (System Block Diagram).

***Figure 1***

***System Block Diagram***




Based on the highly interconnected subsystems, the module can achieve a real-time hazard detection and dynamic alerting.

**ii. Hardware Selection and Decision Matrices**

The three functional groups in the hardware architecture of this module are the central microcontroller, sensing inputs and feedback outputs. The above modules work together to achieve multi-directional detection and prompt response functions. Quantitative decision matrices were used to evaluate and select the best components for the final prototype.

By using this method, the Arduino Uno R3 was selected as the central microcontroller. The following tables illustrate the detailed comparisons among three

main competitors: Arduino Uno R3, Raspberry Pi, and ESP32.

***Table 1***

***Microcontroller Technical Specifications Comparison***

| Specification/ Feature | Arduino Uno R3 (Selected) | Raspberry Pi 4 (Microcomputer) | ESP32 Devkit (Planned for V3.0) |
|---|---|---|---|
| Clock Speed/ Cores | 16MHz (Single Core) | 1.5Ghz (Quad Core) | 240 MHz (Dual Core) |
| Operating Voltage | 5V | 5V | 3.3V |
| Average Power Draw | ~50 mA (Low) | ~600 mA (Very High) | ~80 mA (Moderate, high in Wi-Fi) |
| Weight (Board Only) | 25g | 45g | 10g |
| Estimated Cost | $15 - $25 | $45 - $75 | $5 - $10 |
| Connectivity | None (Wired Only) | Wi-Fi, Bluetooth, Ethernet | Wi-Fi, Bluetooth |
| System Complexity | Low (Bare-metal RTOS) | High (Requires Linux OS) | Medium (RTOS support) |

***Table 2***

***Quantitative Decision Matrix for Central Processing Unit***

| Evaluation Criteria | Weight | Arduino Uno R3 | Raspberry Pi 4 | ESP32 Devkit |
|---|---|---|---|---|
| Power Efficiency | 0.25 | 5 | 1 | 4 |
| Weight and Form Factor | 0.25 | 3 | 1 | 5 |
| Cost Effectiveness | 0.20 | 4 | 1 | 5 |
| Processing Power | 0.15 | 2 | 5 | 4 |
| 5V Sensor Compatibility and Ease of Prototyping | 0.15 | 5 | 2 | 3 |
| Total Weighted Score | 1.00 | 3.85 (Winner for Prototype) | 1.75 (Rejected) | 4.30 (Winner for Future Improvements) |

As shown in the decision matrix (Table 2), the Raspberry Pi 4 has a good processing capacity; however, due to its high-power consumption (600mA) and the

requirement for a powerful external power bank, it was excluded early in the design stage because it exceeded the 150g weight limit for the cane's natural sweeping device.

While the ESP32 scored the highest overall for its compact size and excellent wireless capabilities, its native 3.3V logic level requires additional level-shifting circuitry to interface with the chosen 5V ultrasonic and infrared sensors. Therefore, for the initial prototype, the Arduino Uno R3 was selected. The Arduino Uno R3 provides perfect 5V compatibility, sufficient processing power through algorithmic improvement, and extreme power efficiency. In this way, it is the most practical choice for prototyping. At the same time, the ESP32 remains the optimal choice for future PCB integration.

A decision matrix for obstacle detection was introduced to select a suitable sensor for mid-air hazard detection (see Table 1). Sonar (Ultrasonic) and Infrared (IR) were the two candidates, and their three main evaluation criteria were ease of use, cost-effectiveness and the ability to detect dark or light-absorbing materials, rated from 1 to 5.

***Table 3***

***Obstacle Detection Decision Matrix***

| Criteria (Scale 1-5) | Sonar (Ultrasonic) | Infrared (IR) |
|---|---|---|
| Easy to Use | 5 | 5 |
| Low Cost | 5 | 4 |
| Can Detect Black Easily | 5 | 2 |
| Total Score | 15 | 11 |

As shown in Table 3, the ultrasonic sensor (sonar) had the highest overall score because it was relatively reliable in detecting dark or light-absorbing materials and cost-effective. Therefore, it was chosen as the main mode for the forward-facing mid-air obstacle detection subsystem. However, in order to create an all-encompassing mobility aid, many environmental problems must be solved first. Although an ultrasonic sensor can detect a forward obstacle, it is not ideal for detecting a sudden drop-off at ground level because of a wide beam angle and the phenomenon of sound wave scattering (specular reflection) when directed downwards. Although the infrared proximity sensor scored lower in the main matrix, it is still very good for short-range and fixed-angle proximity detection.

An additional dual-sensor structure has been added to the final system for environmental awareness. The selected ultrasonic sensor will be used as the main forward obstacle detection module, and a downward-facing infrared sensor will be

added as a secondary module for drop-off detection. Sensor fusion will be used to combine the strengths of both technologies and improve user safety.

***Table 4***

***User Feedback Decision Matrix***

| Criteria (Scale 1-5) | Voice (Audio) | Pressure (Vibration) |
|---|---|---|
| Easy to Use | 5 | 3 |
| Quick Response | 3 | 5 |
| Total Score | 8 | 8 |

Table 4 indicates that while Voice feedback is easy to understand, it lacks the immediate reaction urgency required for physical hazards. From the matrix, both audio and haptic feedback system show advantages. In this way, rather than selecting a single output, the device integrated both mechanisms to create a Dual Progressive Feedback System.

For the auditory component, the buzzer acts as a dynamic indicator. Taking advantage of the human auditory system's high sensitivity to rhythm, the software logic dynamically increases frequency inversely with the obstacle's distance, which is similar to automotive parking sensors. At the same time, the vibration motor provides continuous proportional haptic feedback. By utilizing Pulse Width Modulation (PWM) mapping, the physical intensity of the vibration increases linearly as the user approaches a hazard. This dual-modality system ensures that the user receives a high-resolution sensing of their surroundings.

In addition, Light (LED) scored definitely the lowest for blind user feedback but was repurposed in our design exclusively as an external warning signal for surrounding pedestrians. To be more specific, it is designed to alert the surrounding individuals and make them be aware of the visually impaired in the dark environment.

Table 5 describe the circuit connection and the function of each component. Please refer to Appendix C for the integral electric circuit layout for the VisionCane system.

***Table 5***

***Hardware Connection***

| Component | Arduino Pin | Signal Type | Description |
|---|---|---|---|
| Ultrasonic Sensor (HC-SR04) | Trig→D9<br>Echo→D10 | Digital Out/ In | Trig emits ultrasonic pulses, and Echo receives the return echo for calculation. |
| Infrared Sensor | Signal→A0 | Analog Input | Reads continuous analog voltage from surface reflections, enabling dynamic baseline calibration and drop-offs detection |
| Photoresistor | Signal→A0 | Analog Input | Captures ambient light variations to determine if the |

| | | | |
|---|---|---|---|
| | | | environment crosses the dark threshold |
| Vibration Motor | Signal→D3 | PWM Output | Provide proportional haptic feedback based on obstacle proximity |
| Buzzer | Signal→D5 | Digital Output | Provide intermittent or continuous acoustic alerts |

**iii. Software and Program Design**

The software architecture was developed using the Arduino IDE (C++ based) and pre-validated through the Tinkercad circuit simulation environment.

There were two main stages developed in the software design process. The design achieved 5 main improvements from Version 1.0 to Version 2.0.

***a. Non-blocking Polling Architecture and Continuous Proportional Mapping Algorithm***

In the initial prototype (Version 1.0), the control logic relied on the standard delay functions. As the Arduino Uno operates on a single-core microcontroller, it executes only one instruction at a given moment. In this way, this traditional approach introduced blocking periods where the microcontroller was paused, leading to slight delays in sensor polling. To solve the limitations, the final design (Version 2.0) introduces an algorithmic innovation of Non-blocking Polling Architecture, using millis () timer in the microcontroller. By tracking timestamps instead of forcing the

microcontroller to wait, the main execution loop remains unblocked, dramatically reducing system latency to under 50 milliseconds and allowing for true simultaneous multi-tasking of the sensor array (high-speed concurrency rather than true parallel processing).

Additionally, the final design introduces an algorithmic innovation to improve the feedback alerting. In the initial design, the device provided constant haptic and auditorial feedback, lacking the reflection of individual's actual movement. In this way, the users can only notice that they are in the danger zone without knowing how close to the obstacles they are. In the final design, I set a Dual Progressive Feedback System. Specifically, the feedback system does not remain stagnant. The intensity of both the haptic and auditorial feedback will dynamically and smoothly increase as the users are approaching the obstacles. This demonstrates that improvement in high-resolution spatial awareness and sensory preservation can be realized solely through software innovation without much change of the hardware.

***b. Dynamic Baseline Calibration Algorithm***

In terms of the function of drop-offs detection achieved by the IR sensor, the module introduces the Dynamic Baseline Calibration in Version 2.0. The traditional kind of infrared proximity detection is relatively sensitive to changes in the environment. IR sensors are based on light reflection, so different surface materials will produce different amounts of light reflection and thus different analogue signals when exposed to the same light environment on a flat surface. Thus, if a fixed threshold is used for drop-off detection, the precision of detection will be relatively

low when users are moving between different floor areas or in certain light conditions.

***c. Moving Average Filter Algorithm***

The first problem of ultrasonic spatial sensing is that it is easily influenced by transient acoustic interference and specular reflection, and thus produces sudden and incorrect distance measurements. In this way, any inaccurate distance reading will lead to severe problem of sudden alerting. During early testing, relying on raw, single-point data caused the system to produce a high rate of false haptic and auditory alarms. A Moving Average Filter was used in the Version 2.0 software architecture to reduce this sensory unreliability.

A circular buffer array is employed in the Moving Average Filter algorithm to store the most recent distance measurements. The distance output is dynamically and continuously calculated according to the model below:

$$D_{average} = \frac{1}{N}\sum_{i=1}^{N} D_i$$

Where N=5 represents the predefined buffer capacity, and $D_i$ refers to the historical raw distance samples. With this calculation model, as each new measurement is acquired, the oldest data will be overwritten.

This algorithm can effectively improve the accuracy of the distance output for further alerting feedback. Under the framework, an isolated and abnormal reading result is mathematically diluted by its surrounding stable data, preventing inaccurate reading reports.

Additionally, the buffer size of N=5 is not selected arbitrarily but derived from a serious mathematical analysis. The non-blocking execution interval is set at 50 milliseconds. Therefore, a buffer size of 5 introduces a maximum verification latency of 250 milliseconds ($5 \times 50$ ms). Considering that the visually impaired typically navigates at an average walking speed of approximately 1.2m/s according to the standard recommended by the Manual on Uniform Traffic Control Devices (MUTCD), the user would travel a maximum distance of 30 cm during the buffer window. As this 30 cm travel distance is strictly within the 50 cm Danger Zone threshold, N=5 represents the optimal engineering design. In this way, it can guarantee a highly precise distance output without sacrificing the safety buffer zone for the users.

***d. Environmental Adaptive Learning***

To address the high rate of incorrect detection caused by varying surface reflectivities and environmental light conditions, an Environmental Adaptive Learning is introduced. In this way, rather than relying on a static boot-time calibration, the system employs a Leaky Integrator. If a brief anomaly occurs, the integrator slowly leaks out the outdated history and absorbs the new reading from reality. To make it more acceptable, let' s assume that there is a bottle of pure water. Every time we detect a drip of water with another color, we will pour out the same amount of water from the original bottle and add the new drip of water with other color to the original bottle. In this way, the mixture of original pure water and the new added water will be our final result. The adaption logic is mathematically modeled as follow:

$$\boldsymbol{B_{new} = \alpha \cdot B_{old} + (1 - \alpha) \cdot V_{current}}$$

where B represents the baseline, $V_{current}$ is the real-time sensor reading, and the decay factor is set to $\alpha = 0.99$. By assigning 99% computational weight to historical data and 1% to the current reading, the algorithm filters out high-frequency transient noise while smoothly calibrating itself to shifts in the environments. The decay factor $\alpha$ is not selected randomly. The following is the strict mathematical deduction. By setting $\alpha = 0.99$, the time constant $\tau \approx \frac{T_s}{1-\alpha}$ of the Leaky Integrator is approximate to 5 seconds. This time constant is chosen to correspond with human walking mechanics: it is longer than the transient perturbations caused by sweeping over different surfaces (<1s). In this way, it allows the system to adapt smoothly when transitioning between different surfaces.

## V. Verifications and Results

### i. Component and Subsystem Testing

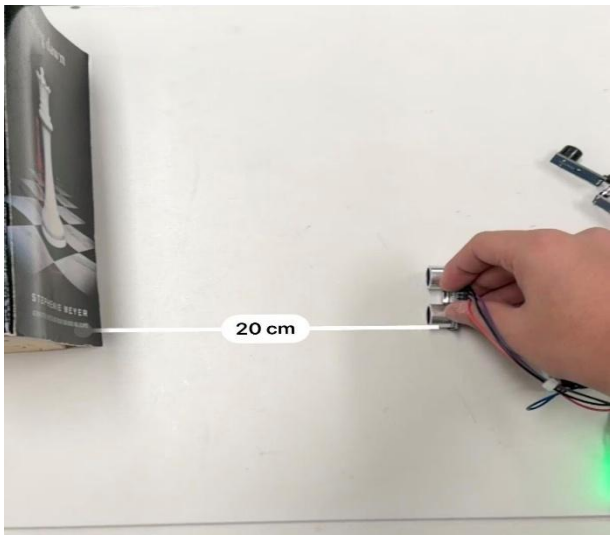
In terms of the testing of components of the CaneVision, I focus on verifying these following parts: Ultrasonic Sensor, Infrared Sensor and Photoresistor with LED.

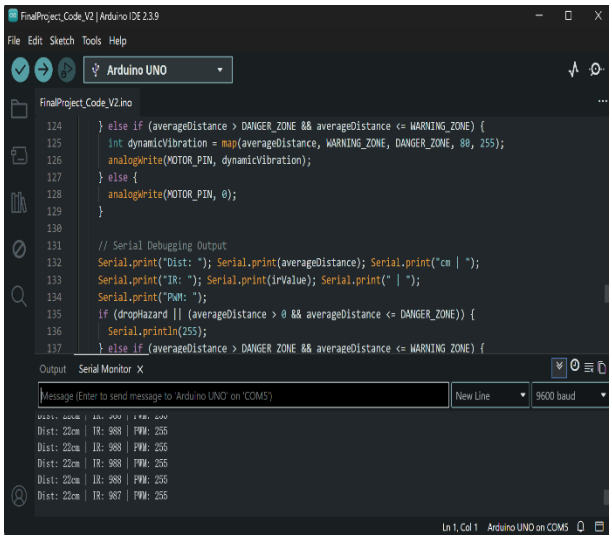
#### *a. Ultrasonic Sensor*

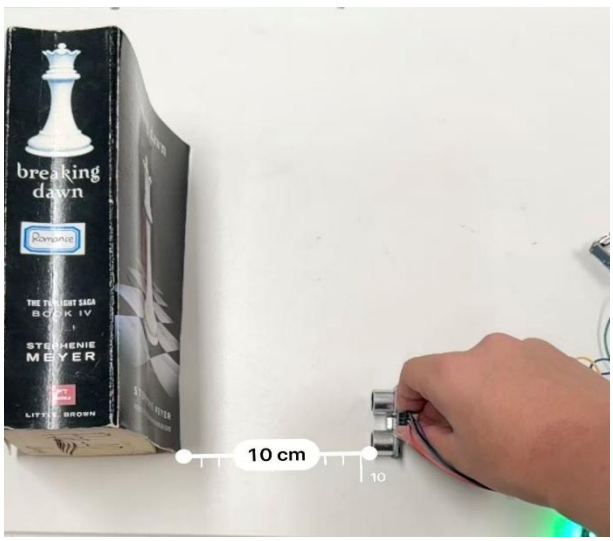
To establish a reliable baseline, the HC-SR04 ultrasonic sensor was evaluated for its static ranging accuracy. Empirical testing involved placing a flat physical barrier at known intervals: 10cm, 20cm, and 30cm. The results are shown below:

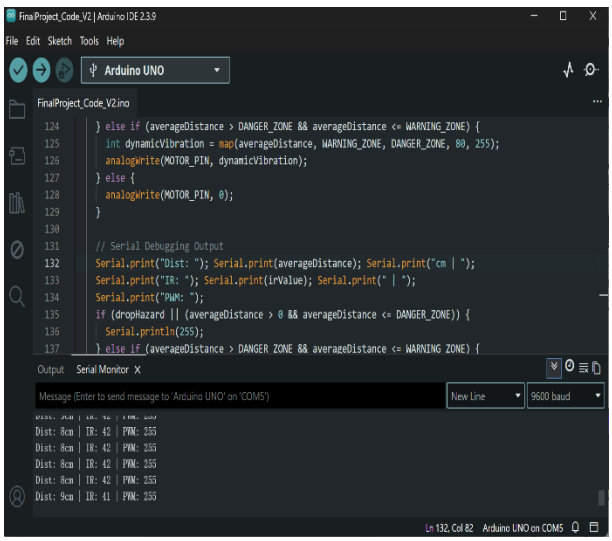
***Figure 3***

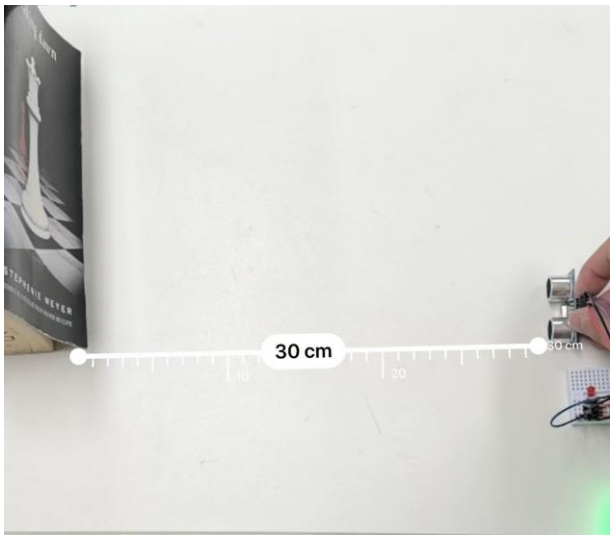
***Ultrasonic Sensor Testing Result***





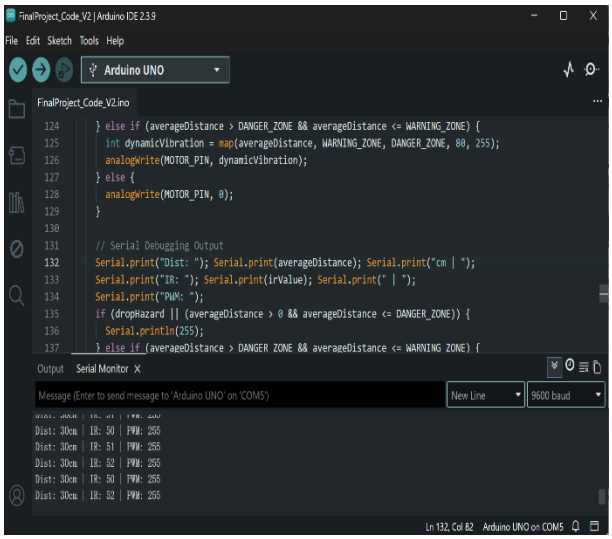




***Table 6***

***Ultrasonic Testing Result***

| Measured Distance | Actual Distance | Error |
|---|---|---|
| 8cm | 10 cm | 20% |
| 22cm | 20 cm | 10% |
| 30cm | 30 cm | 0% |

*Note*. The physical measurements are shown on the left, while the serial monitor outputs are shown on the right.

The output of the serial monitor showed a relatively small error margin of less than $\pm 2$ cm and was suitable for general-purpose pedestrian navigation.

***b. Infrared Sensor***

The basic physical response characteristics of a downward-facing infrared sensor were determined. In the static test, the raw analogue output at a normal flat surface was about 50. When the module was moved outside the edge of the test platform to simulate a sudden drop-off, the analog reading immediately and significantly

increased to more than 600. Therefore, it was determined that the hardware could not detect a ground-potential condition before any elaborate processing. The experimental IR value and the adopted baseline are shown in Table 5 below.

***Table 7***

***Infrared Sensor Dynamic Baseline and Drop-off Detection Validation***

| Test Surface | Raw IR Value | Current Baseline |
|---|---|---|
| Flat White Surface (Calibration) | 52 | 52 |
| Flat Dark Surface (Adaptive) | 105 | 105 (Adapted over 5s) |
| Table Edge (Transitional) | 280 | 105 (Ignored by filter) |
| Mid-Air/ Drop-off | 640 | 105 (Ignored by filter) |

***c. Photoresistor and Actuator***

The photoresistor has been tested in various illumination conditions. The results reveal that the photoresistor is able to identify specific illumination circumstances and trigger the LED to be on or off based on its reading.

At the same time, the independent output devices of vibration motors and piezo buzzers were also tested for hardware response. During the tests, the buzzer (auditory actuator) worked correctly when given the two-state binary logic signals (HIGH/LOW). A variable Pulse Width Modulation (PWM) experiment was carried out to test the proportional Haptic Feedback system. The Arduino R3 was programmed to output a specific PWM value that represents the distance of a virtual obstacle. A digital multimeter was used to measure the mean voltage at the motor terminals, and the intensity of physical vibration was qualitatively recorded (as shown in Table 8). Based on the above experiments, haptic feedback is enhanced linearly and perceptibly when the obstacle exceeds the 150 cm warning threshold and approaches the 50 cm critical danger zone.

***Table 8***

***Haptic Feedback Validation***

| Simulated Distance (cm) | PWM Command (0-255) | Average Measure Voltage | Motor Status/ Perceived Intensity |
|---|---|---|---|
| >150 (Safe Zone) | 0 | 0.00 V | OFF (No Vibration) |

| | | | |
|---|---|---|---|
| 120 (Warning Zone) | 132 | 2.60 V | Weak, noticeable pulsing |
| 80 (Warning Zone) | 202 | 3.96 V | Moderate, continuous vibration |
| <=50 (Danger Zone) | 255 | 5.00 V | Maximum intensity vibration |

**ii. Subsystem Testing**

***a. Moving Average Filter System***

A subsystem of a Moving Average Filter was tested by adding transient noise to the measurement. Although there were some high-frequency spikes (erroneous readings exceeding 300 cm) in the raw sensor data, the filtered output successfully reduced these fluctuations. Therefore, the system will maintain the stability of the haptic and audio feedback.

***b. Environmental Adaptive System***

Being one of the most crucial parts of my final design, the subsystem of Environmental Adaptive System was validated by setting the device in various conditions. When the device was transitioned from a highly reflective surface to a dark, light-absorbing material, the system did not trigger a false drop-off alarm. Serial results showed that the dynamic baseline smoothly adjusted to the new environment over a period of about 5 seconds.

**c. Full System and Functional Integration Test**

All three were used simultaneously in the dynamic simulated navigation: ultrasonic sensor, infrared sensor and photoresistor. Based on observation, the microcontroller can perform several tasks concurrently at a high speed. The total system latency from physical obstacle detection to the beginning of mechanical feedback is less than 100ms, as required. In this way, the device as an integration of different subsystems ensures real-time responsiveness for a user walking at standard speeds.

To evaluate the efficacy of the Dual Progressive Feedback System, a specific scenario with multiple obstacles was conducted. As the device approached a physical barrier, the haptic and auditory outputs correctly initiated the Warning Zone by providing intermittent feedback at the 150 cm threshold. As the distance decreased, the pulse frequency increased proportionally, revealing that the users transitioned into the Danger Zone.

In addition, an edge-case override test was also performed by simulating a forward obstacle and a sudden ground-level drop-off. The software logic correctly recognized a serious hazard. After detecting the drop-off, the system will trigger a maximum haptic and auditory alarm. Thus, the integration logic effectively protected the user in a complex, multi-variable environment without crashing the system.

## VI. Conclusions and Recommendations

### i. Conclusions

The CaneVision project successfully demonstrated that advanced and multi-direction obstacle perception can be integrated into the traditional white cane without sacrificing affordability or the user's natural mobility mechanics. A sensor fusion structure will be adopted to combine ultrasonic detection of the upper-body obstacle and infrared sensor detection of a ground-level drop-off to compensate for the blind spots of traditional white canes.

Crucially, the project solved the problem of widespread false-positive alarms caused by transient noise and various surface reflections through the application of software innovation. A Moving Average Filter and an Environmental Adaptation System were added to address the problem of inaccurate detection results caused by the hardware. For the user-centered design criteria, the Dual Progressive Feedback System effectively addressed the sensory overload problem common in commercial aid device. By prioritizing haptic feedback and restricting the auditory buzzer for Danger Zone(<50cm), the CaneVision design successfully protects the user's natural awareness of the environment.

Last but not least, the final design keeps the total material cost under $50 threshold (refer to Appendix B), meeting the economic constraint. For this reason, we are convinced that the high-resolution aid device can be highly accessible to the broader visually impaired community.

### ii. Recommendations for Future Improvements

While the current CaneVision prototype effectively meets the initial design criteria, several strategies are recommended for future improvements:

#### *a. Advanced Power Management*

For extended daily operation of the system, a standalone Lithium-Polymer (Li-Po) battery module and a TP4056 charging circuit need to be added. Additionally, a "short break" mode can be selected for a stationary cane to save more battery power.

***b. Hardware Miniaturization and Custom PCB***

The current Arduino Uno development board is relatively bulky. Future design should transition to a custom-designed Printed Circuit Board (PCB) utilizing a surface-mount microcontroller (e.g., ESP32). It will be smaller in size, and the whole module can fit inside the handle of the cane.

***c. Protective Housing***

The physical enclosure should be upgraded using SLA 3D printing technology to create a protective housing for various weather and conditions. A clip-on clamping part should be made, and this part would snap shut around the cane. In this way, the same part could fit cane sticks with different thicknesses, and screws would not be needed at all.

## VII. Reference

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# VIII. Appendices

## Appendix A: Complete Source Code

```
/*
 * Project: CaneVision Smart Navigation Module (V3.0)
 * Jingkun Li (Jarred) Carlotta Berry Ph.D. Pioneer Academics
 * Features: Attachable module for the traditional white cane. The whole system involves multi-direction detection system and alerting system.
 * Moving Average Filter;  Non-blocking millis() Architecture; Environment Adaptive Learning
 */

// Define hardware connections for sensors and actuators
const int TRIG_PIN = 9;
const int ECHO_PIN = 10;
const int IR_PIN = A0;
const int LIGHT_PIN = A1;
const int MOTOR_PIN = 3;
const int BUZZER_PIN = 5;
const int LED_PIN = 6;

// Define operational thresholds and zones for spatial awareness
const int WARNING_ZONE = 150;
const int DANGER_ZONE = 50;
const int DARK_THRES = 300;

// Variables for IR sensor dynamic baseline and adaptive learning
int irBaseline = 0;
int dropThreshold = 0;
const int CALIBRATION_SAMPLES = 50;
const int DROP_OFFSET = 150;

// Variables for ultrasonic moving average filter
const int NUM_READINGS = 5;
int readings [NUM_READINGS];
int readIndex = 0;
int total = 0;
int averageDistance = 0;

// Variables for non-blocking timing architecture
unsigned long previousMillis = 0;
const long loopInterval = 50;

// Variables for dynamic auditory feedback (Non-blocking buzzer)
```

```
unsigned long previousBuzzerMillis = 0;
int buzzerState = HIGH;

void setup () {
  Serial.begin(9600);

  pinMode (TRIG_PIN, OUTPUT);
  pinMode (ECHO_PIN, INPUT);
  pinMode (IR_PIN, INPUT);
  pinMode (LIGHT_PIN, INPUT);
  pinMode (MOTOR_PIN, OUTPUT);
  pinMode (BUZZER_PIN, OUTPUT);
  pinMode (LED_PIN, OUTPUT);

  // Initialize outputs safely (HIGH means OFF for this active buzzer)
  analogWrite (MOTOR_PIN, 0);
  digitalWrite (BUZZER_PIN, HIGH);
  digitalWrite (LED_PIN, LOW);

  // --- Ultrasonic Sensor Initialization Fix (Prevent Cold Start Bug) ---
  // Get an initial real distance reading to populate the filter array
  digitalWrite (TRIG_PIN, LOW);
  delayMicroseconds (2);
  digitalWrite (TRIG_PIN, HIGH);
  delayMicroseconds (10);
  digitalWrite (TRIG_PIN, LOW);
  long initDuration = pulseIn (ECHO_PIN, HIGH);
  int initDistance = initDuration * 0.034 / 2;

  // Fill the array with the real distance to avoid false low averages
  for (int i = 0; i < NUM_READINGS; i++) {
    readings[i] = initDistance;
    total += initDistance;
  }


  Serial.println("Calibrating IR Baseline... Keep the cane steady on flat ground.");
  digitalWrite(LED_PIN, HIGH);

  long tempSum = 0;
  for(int i = 0; i < CALIBRATION_SAMPLES; i++) {
    tempSum += analogRead(IR_PIN);
    delay(50);
  }
```

```
  irBaseline = tempSum / CALIBRATION_SAMPLES;
  dropThreshold = irBaseline + DROP_OFFSET;

  digitalWrite(LED_PIN, LOW);
  Serial.print("Calibration Complete! Drop Threshold set to: ");
  Serial.println(dropThreshold);
  delay(1000);
}

void loop() {
  unsigned long currentMillis = millis();

  //Execute main logic every 50ms
  if (currentMillis - previousMillis >= loopInterval) {
    previousMillis = currentMillis;

    // Ambient light evaluation
    int lightLevel = analogRead(LIGHT_PIN);
    digitalWrite(LED_PIN, (lightLevel > DARK_THRES) ? HIGH : LOW);

    // Ultrasonic ranging via pulse generation
    digitalWrite(TRIG_PIN, LOW);
    delayMicroseconds(2);
    digitalWrite(TRIG_PIN, HIGH);
    delayMicroseconds(10);
    digitalWrite(TRIG_PIN, LOW);

    long duration = pulseIn(ECHO_PIN, HIGH);
    int currentDistance = duration * 0.034 / 2;

    // Moving Average Filter execution
    total = total - readings[readIndex];
    readings[readIndex] = currentDistance;
    total = total + readings[readIndex];
    readIndex = (readIndex + 1) % NUM_READINGS;
    averageDistance = total / NUM_READINGS;

    // IR surface adaptation and hazard detection
    int irValue = analogRead(IR_PIN);
    bool dropHazard = (irValue > dropThreshold);

    if (!dropHazard) {
      irBaseline = (irBaseline * 99 + irValue) / 100;
```

```
      dropThreshold = irBaseline + DROP_OFFSET;
    }

    // Proportional Feedback Logic (Motor Control)
    if (dropHazard || (averageDistance > 0 && averageDistance <= DANGER_ZONE)) {
      analogWrite(MOTOR_PIN, 255);
    } else if (averageDistance > DANGER_ZONE && averageDistance <= WARNING_ZONE) {
      int dynamicVibration = map(averageDistance, WARNING_ZONE, DANGER_ZONE, 80,
255);
      analogWrite(MOTOR_PIN, dynamicVibration);
    } else {
      analogWrite(MOTOR_PIN, 0);
    }

    // Serial Debugging Output
    Serial.print("Dist: "); Serial.print(averageDistance); Serial.print("cm | ");
    Serial.print("IR: "); Serial.print(irValue); Serial.print(" | ");
    Serial.print("PWM: ");
    if (dropHazard || (averageDistance > 0 && averageDistance <= DANGER_ZONE)) {
      Serial.println(255);
    } else if (averageDistance > DANGER_ZONE && averageDistance <= WARNING_ZONE) {
      Serial.println(map(averageDistance, WARNING_ZONE, DANGER_ZONE, 80, 255));
    } else {
      Serial.println(0);
    }
  }

  // Independent Non-blocking Auditory Feedback Architecture
  // This runs continuously outside the 50ms loop to ensure precise acoustic timing
  int irValueAsync = analogRead(IR_PIN);
  bool dropHazardAsync = (irValueAsync > dropThreshold);

  if (dropHazardAsync || (averageDistance > 0 && averageDistance <= DANGER_ZONE)) {

    // Critical Danger: Solid continuous beep
    digitalWrite(BUZZER_PIN, LOW);

  } else if (averageDistance > DANGER_ZONE && averageDistance <= WARNING_ZONE) {

    // Warning Zone: Dynamic beep intervals based on proximity
    // Maps distance (50-150cm) to beep interval (100ms to 800ms)
    int beepInterval = map(averageDistance, DANGER_ZONE, WARNING_ZONE, 100, 800);

    if (currentMillis - previousBuzzerMillis >= beepInterval) {
```

```
      previousBuzzerMillis = currentMillis;

      // Toggle the buzzer state to create a pulsing beep
      buzzerState = (buzzerState == HIGH) ? LOW : HIGH;
      digitalWrite(BUZZER_PIN, buzzerState);
    }

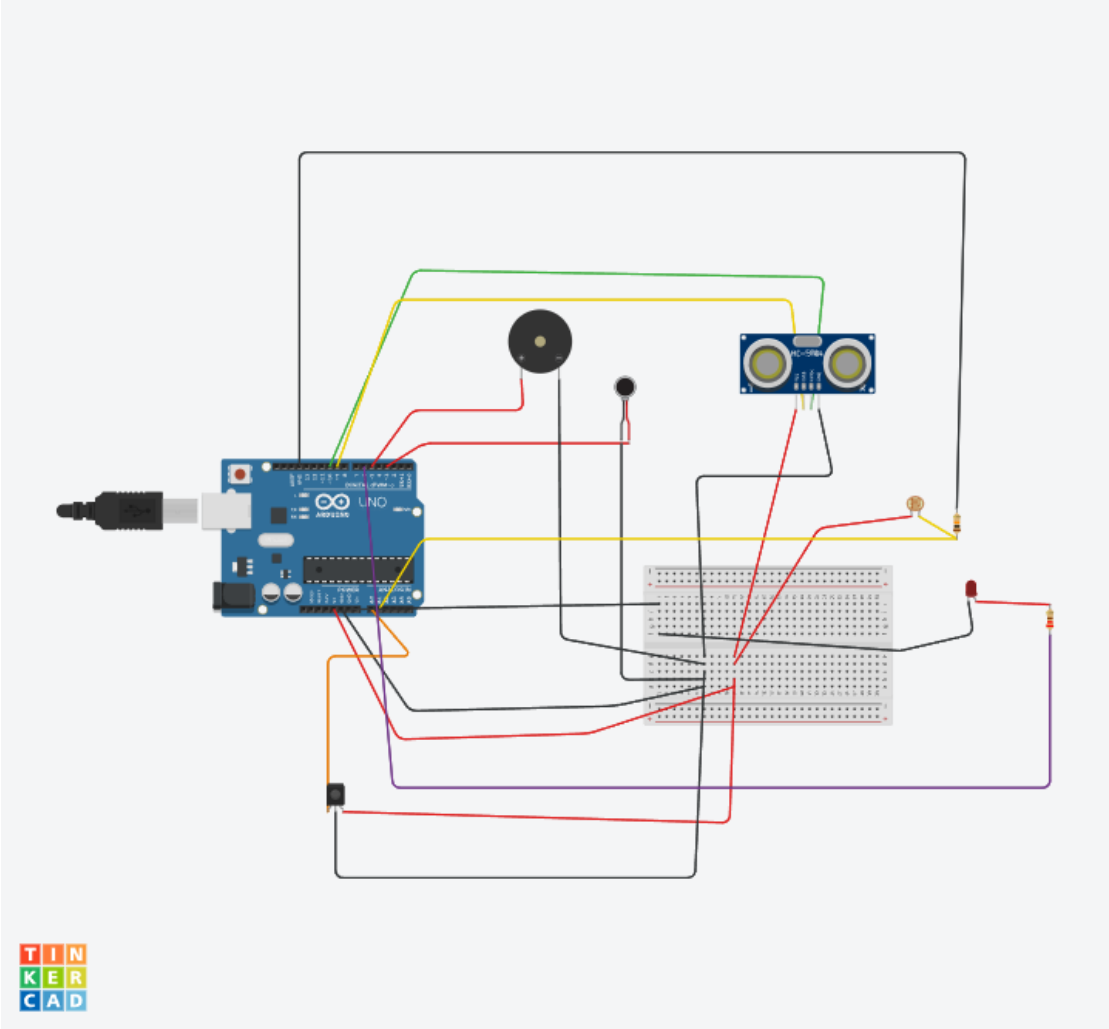
  } else {

    // Safe Zone: Keep buzzer completely OFF
    digitalWrite(BUZZER_PIN, HIGH);
    buzzerState = HIGH;
  }
}
```

## Appendix B: Bill of Materials (BOM)

| Part Description | Quantity | Link | Cost | Notes |
|---|---|---|---|---|
| Arduino Uno | 1 | https://www.adafruit.com | $13.50 | Controller |
| Ultrasonic Sensor | 1 | https://www.adafruit.com/product/4007 | $3.95 | Detection |
| Vibrating Mini Motor Disc | 1 | https://www.adafruit.com/product/1201 | $1.95 | Avoidance |
| Buzzer | 1 | https://www.adafruit.com/product/160 | $1.5 | Avoidance |
| Breadboard | 1 | https://www.adafruit.com/product/64 | $4.95 | / |
| IR sensor | 1 | https://www.adafruit.com/product/164 | $14.95 | Detection |
| Photoresistor | 1 | https://www.adafruit.com/product/161 | $0.95 | Detection |
| LED | 1 | https://www.adafruit.com/product/2700 | $0.19 | Avoidance |
| Total | 8 | / | $41.94 | / |

## Appendix C: Circuit Schematic and Pin Mapping



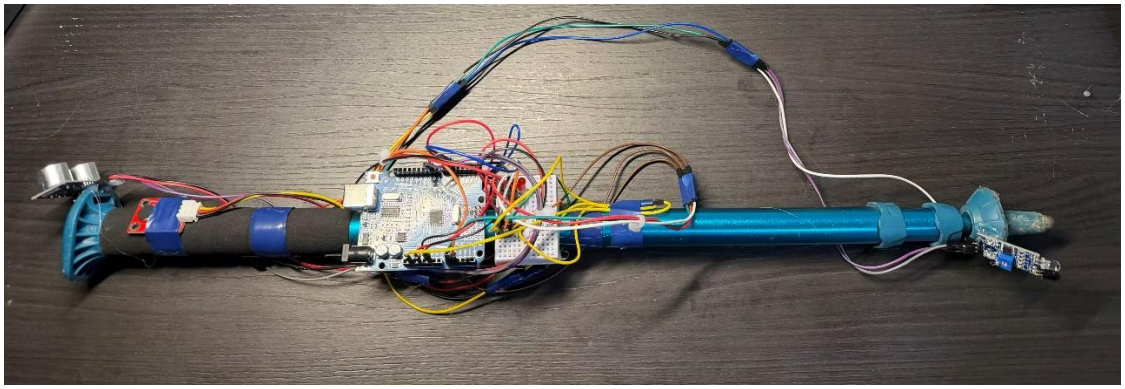
*Circuit Layout*

| Component | Arduino Pin | Signal Type | Description |
|---|---|---|---|
| Ultrasonic Sensor (HC-SR04) | Trig→D9<br>Echo→D10 | Digital Out/ In | Trig emits ultrasonic pulses, and Echo receives the return echo for calculation. |
| Infrared Sensor | Signal→A0 | Analog Input | Reads continuous analog voltage from surface reflections, enabling dynamic |

| | | | |
|---|---|---|---|
| | | | baseline calibration and drop-offs detection |
| Photoresistor | Signal→A0 | Analog Input | Captures ambient light variations to determine if the environment crosses the dark threshold |
| Vibration Motor | Signal→D3 | PWM Output | Provide proportional haptic feedback based on obstacle proximity |
| Buzzer | Signal→D5 | Digital Output | Provide intermittent or continuous acoustic alerts |

*Circuit Connection*

**Appendix D: Final Assembled Hardware and Live Demonstration**



*Final Assembled Hardware*

| | |
|---|---|
| | Video Title: CaneVision Desktop Demo Presentation<br><br>Access Link:<br>https://youtube.com/shorts/EYo3PqKyFFs?feature=share<br><br>*Note: Scan the QR code or click the link above to view the live demonstration video.* |